Quiz III

Name: **Entry Number:**

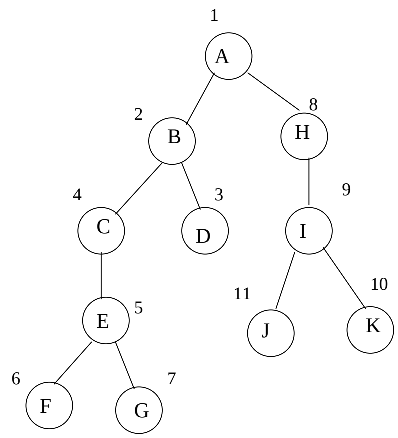
1. (6 marks) Let V be a vector subspace of $\Re^n$. Let W be the set of vectors $\{x : \langle x, v\rangle = 0, \text{ for all } v \in V\}$. Show that W is a subspace. Show that every vector $x \in \Re^n$ can be uniquely written as $x_1 + x_2$ where $x_1 \in V, x_2 \in W$.

2. Write down the singular value decompposition of the following matrices:

(a) (3 marks) $\begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$

(b) (3 marks) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

3. (3 marks) Consider the following DFS tree of a directed graph. On top of each node, a number is written this number denotes the order in which the nodes were visited (i.e., the increasing order of discovery times). Which of the following edges cannot be there in the graph: (A,C), (C,A), (C,J), (C,D), (E,I), (K,B) ? Give reasons.



4. (5 marks) Consider the undirected graph below. We perform BFS (Breadth First Search) starting from vertex 1. Assume that we use adjacency list data-structure and each vertex maintains the list of neighbours in decreasing order of their indices (e.g., vertex 1 maintains the list in the order 14, 7, 6, 2). Print the order in which the vertices will be added to the queue. Further, draw the resulting BFS tree.